

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		catalyst in a different physical state from the reactants / reaction	1			1		
		(ii)	I	$[2 \times \Delta_f H(\text{SO}_3)] - [2 \times \Delta_f H(\text{SO}_2)] = -197$ (1) $\Delta_f H(\text{SO}_3) = -395.5 \text{ kJ mol}^{-1}$ (1)		2		2	2	
			II	$T = \frac{\Delta H}{\Delta S}$ (1) $T = 1053 \text{ K}$ (1) student is incorrect must give reason ΔG is negative below this temperature so the reaction is feasible at a lower temperature (1)		2	1	3	2	
			III	initial concentrations of SO_2 and $\text{O}_2 = 0.020 \text{ mol dm}^{-3}$ (1) equilibrium concentrations – both needed for (1) $[\text{SO}_2] = 0.006 \text{ mol dm}^{-3}$ $[\text{O}_2] = 0.013 \text{ mol dm}^{-3}$ $K_c = \frac{(0.014)^2}{(0.006)^2 \times (0.013)} = 419$ (1) $\text{mol}^{-1} \text{ dm}^3$ (1)		4		4	4	
				Question 8 total	5	12	1	18	8	0